

```

//Program to find the sum of all prime numbers between 1 and 1000.
//Prime numbers are those numbers which are not divisible by any number other than 1 and themselves.
//The sum of prime numbers between 1 and 1000 is 13814.

//Function to check if a number is prime or not.
bool isPrime(int n)
{
    if (n < 2)
        return false;
    for (int i = 2; i <= sqrt(n); i++)
        if (n % i == 0)
            return false;
    return true;
}

//Function to find the sum of all prime numbers between 1 and 1000.
int sumOfPrimes(int n)
{
    int sum = 0;
    for (int i = 2; i <= n; i++)
        if (isPrime(i))
            sum += i;
    return sum;
}

//Main function
int main()
{
    int n = 1000;
    int sum = sumOfPrimes(n);
    cout << "Sum of prime numbers between 1 and 1000 is: " << sum << endl;
    return 0;
}

//Output
Sum of prime numbers between 1 and 1000 is: 13814

//Complexity
//Time Complexity: O(n * sqrt(n))
//Space Complexity: O(1)

//Note: The sum of prime numbers between 1 and 1000 is 13814.
```



Prove that the sum of the lengths of the sides of a triangle is less than the sum of the lengths of the medians of the triangle.

The following is the proof to establish the above property.

```

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